Lecture 26

Orthogonal branching and classical comparisons

Objectives. *We restrict orthogonal modules along the chain $\mathfrak{so}_{2n+1} \supset \mathfrak{so}_{2n} \supset \mathfrak{so}_{2n-1}$. The highest weights interlace. The spin cases are proved in the exterior model. A symplectic comparison shows why an embedding must be named first.*

By the end of this lecture you will be able to:

- Apply the orthogonal interlacing rules to weights of integer and half-integer type.
- Prove from the exterior model how spin and half-spin modules restrict.
- Specify an embedding before decomposing a restriction to a root $\mathfrak{sl}_2$.
- Use the cited symplectic branching theorem to find a branching multiplicity greater than one.

26.1 Embeddings and restricted weights

A *branching rule* describes how an irreducible module decomposes on restriction to a subalgebra. The answer depends on the embedding and on how the Cartan subalgebras are matched. We fix both first.

All modules are finite-dimensional over $\mathbb{C}$. Let $n \geq 2$ and let $V = \mathbb{C}^{2n+1}$ have basis $(e_1, \dots, e_n, f_1, \dots, f_n, e_0)$ and the symmetric bilinear form q with

$$q(e_i, f_j) = \delta_{ij}, \qquad q(e_0, e_0) = 1,$$

all other pairings of basis vectors being zero. Let W be the span of the e_i and f_i. Then $\mathfrak{so}(V) \cong \mathfrak{so}_{2n+1}$ and $\mathfrak{so}(W) \cong \mathfrak{so}_{2n}$.

For $t \in \mathbb{C}^n$, let $H_t \in \mathfrak{so}(V)$ act by

$$H_t e_i = t_i e_i, \qquad H_t f_i = -t_i f_i, \qquad H_t e_0 = 0.$$

These operators form a Cartan subalgebra $\mathfrak{h}$ of both algebras, as in Lecture 23. Put $\varepsilon_i(H_t) = t_i$, and write $\lambda = (\lambda_1, \dots, \lambda_n)$ for $\sum_i \lambda_i \varepsilon_i$. Use the dominant integral weights of Lecture 23, with the D_2 convention of Lecture 24. For the rank-one target $B_1 = \mathfrak{so}_3$, dominance means $\lambda_1 \in \frac{1}{2}\mathbb{Z}_{\geq 0}$. The Clifford construction and weight computation below also hold for $n = 1$, using the same anticommutation relations. A weight has *integer type* if all coordinates lie in $\mathbb{Z}$, and *half-integer type* if all lie in $\frac{1}{2} + \mathbb{Z}$.

Lemma 26.1 (Stabiliser of a nonisotropic vector). *Let U carry a nondegenerate symmetric form q, and let $u \in U$ satisfy $q(u,u) \neq 0$. Restriction to $u^{\perp}$ is an isomorphism of Lie algebras*

$$\{X \in \mathfrak{so}(U) : Xu = 0\} \longrightarrow \mathfrak{so}(u^{\perp}).$$

Proof. Since $q(u,u) \neq 0$, we have $U = \mathbb{C}u \oplus u^{\perp}$, and q is nondegenerate on $u^{\perp}$. Let $Xu = 0$. For $x \in u^{\perp}$,

$$q(Xx, u) = -q(x, Xu) = 0,$$

so X preserves $u^{\perp}$ and $X|_{u^{\perp}}$ is skew. Restriction to an invariant subspace preserves commutators, so the restriction map is a homomorphism of Lie algebras. Since $Xu = 0$, the restriction determines X.

Conversely, extend $Y \in \mathfrak{so}(u^{\perp})$ by $Yu = 0$. For $x, y \in u^{\perp}$ and $a, b \in \mathbb{C}$,

$$q\big(Y(x+au), y+bu\big) = q(Yx, y) = -q(x, Yy) = -q\big(x+au, Y(y+bu)\big).$$

The first and last steps use $Yu = 0$ and $q(Yx, u) = q(u, Yy) = 0$. So the extension is skew. ∎

Embedding I. Take $U = V$ and $u = e_0$, so $u^{\perp} = W$. Every H_t kills e_0. Both algebras share $\mathfrak{h}$, and weights restrict unchanged. The positive roots $\varepsilon_i \pm \varepsilon_j$ $(i < j)$ of D_n are among those of B_n, which also contain the ε_i.

Embedding II. Take $U = W$ and $u = e_n + f_n$, so $q(u,u) = 2$. A basis of $u^{\perp}$ is

$$e_1, \dots, e_{n-1}, \quad f_1, \dots, f_{n-1}, \quad e_0' = \frac{i}{\sqrt{2}}(e_n - f_n).$$

Since $q(e_n, f_n) = 1$ and $q(e_n, e_n) = q(f_n, f_n) = 0$, the vector e_0' satisfies

$$q(e_0', e_0') = -\frac{1}{2}\, q(e_n - f_n, e_n - f_n) = -\frac{1}{2}\,(-2) = 1.$$

Thus $u^{\perp}$ is a copy of the $(2n-1)$-dimensional model. The operators H_t with $t_n = 0$ kill e_n and f_n, hence u and e_0'. They form the Cartan subalgebra of $\mathfrak{so}(u^{\perp}) \cong \mathfrak{so}_{2n-1}$ in this basis. Weights restrict by

$$(\lambda_1, \dots, \lambda_{n-1}, \lambda_n) \longmapsto (\lambda_1, \dots, \lambda_{n-1}).$$

26.2 The orthogonal interlacing rules

Theorem 26.2 (From B_n to D_n · *[C]*). *Let $n \geq 2$, use Embedding I, and let λ be dominant for B_n. Then*

$$L(\lambda)|_{\mathfrak{so}_{2n}} \cong \bigoplus_{\mu} L(\mu),$$

summed over D_n-dominant μ of the same type as λ with

$$\lambda_1 \geq \mu_1 \geq \lambda_2 \geq \cdots \geq \lambda_{n-1} \geq \mu_{n-1} \geq \lambda_n \geq |\mu_n|.$$

Each such μ occurs exactly once.

Theorem 26.3 (From D_n to B_{n-1} · *[C]*). *Let $n \geq 2$, use Embedding II, and let λ be dominant for D_n. Then $L(\lambda)|_{\mathfrak{so}_{2n-1}}$ is the direct sum of the $L(\mu)$, each once, over B_{n-1}-dominant $\mu = (\mu_1, \dots, \mu_{n-1})$ of the same type with*

$$\lambda_1 \geq \mu_1 \geq \lambda_2 \geq \cdots \geq \lambda_{n-1} \geq \mu_{n-1} \geq |\lambda_n|.$$

Source. We import the Lie algebra forms of Goodman–Wallach, Theorems 8.1.3–8.1.4, §8.1.1, pp. 364–365. Their Spin subgroups use Embeddings I and II. The labels cover all finite-dimensional modules by their Theorem 6.3.6 and Corollary 6.3.7.

The restriction algorithm. Write λ in ε-coordinates. List the μ of the same type satisfying the inequalities. Check the dimensions with the Weyl dimension formula of Lecture 17.

Example 26.4 (A half-integral chain in rank two). For B_2, $\omega_1 = (1,0)$ and $\omega_2 = (\tfrac12,\tfrac12)$. Take $\lambda = \omega_1 + \omega_2 = (\tfrac32,\tfrac12)$. With Weyl vector $\rho_W = (\tfrac32,\tfrac12)$ we get $\lambda + \rho_W = (3,1)$. Each coroot is a positive multiple of its root, so each ratio in the dimension formula is unchanged if the coroot is replaced by the root. The positive roots $\varepsilon_1 - \varepsilon_2$, $\varepsilon_1 + \varepsilon_2$, ε_1, ε_2 give, in this order,

$$\dim L(\lambda) = \frac{2}{1}\cdot\frac{4}{2}\cdot\frac{3}{3/2}\cdot\frac{1}{1/2} = 2\cdot 2\cdot 2\cdot 2 = 16.$$

The rule requires

$$\frac{3}{2} \geq \mu_1 \geq \frac{1}{2} \geq |\mu_2|, \qquad \mu_1,\mu_2 \in \frac{1}{2}+\mathbb{Z}.$$

So $\mu_1 \in \{\tfrac32,\tfrac12\}$ and $\mu_2 = \pm\tfrac12$.

The roots $\varepsilon_1 \mp \varepsilon_2$ of D_2 are orthogonal, so $\mathfrak{so}_4 \cong \mathfrak{sl}_2 \oplus \mathfrak{sl}_2$ and

$$L(\mu) \cong V_{\mu_1-\mu_2} \boxtimes V_{\mu_1+\mu_2}, \qquad \dim L(\mu) = (\mu_1-\mu_2+1)(\mu_1+\mu_2+1).$$

The four summands have dimensions

$$(\tfrac32,\tfrac12):\ V_1\boxtimes V_2,\ 6; \qquad (\tfrac32,-\tfrac12):\ V_2\boxtimes V_1,\ 6;$$
$$(\tfrac12,\tfrac12):\ V_0\boxtimes V_1,\ 2; \qquad (\tfrac12,-\tfrac12):\ V_1\boxtimes V_0,\ 2.$$

The total is 16.

26.3 Spin restrictions from the exterior model

Let $E = \operatorname{span}(e_1,\dots,e_n)$. Use the creation operators μ_i, contractions ι_i with the dual basis vectors e_i^*, and degree parity θ of Lecture 24 on ΛE. Then

$$\iota_i\mu_j + \mu_j\iota_i = \delta_{ij}, \qquad \iota_i\iota_j + \iota_j\iota_i = 0 = \mu_i\mu_j + \mu_j\mu_i.$$

The operator θ anticommutes with every ι_i and μ_i, and $\theta^2 = 1$. Define the linear map $\gamma : V \to \operatorname{End}(\Lambda E)$ by

$$\gamma(e_i) = \sqrt{2}\,\mu_i, \qquad \gamma(f_i) = \sqrt{2}\,\iota_i, \qquad \gamma(e_0) = \theta.$$

This map satisfies the Clifford relation

$$\gamma(x)\gamma(y) + \gamma(y)\gamma(x) = 2q(x,y) \quad \text{for all } x,y \in V.$$

For example, the anticommutation relations give

$$\gamma(e_i)\gamma(f_i) + \gamma(f_i)\gamma(e_i) = 2(\mu_i\iota_i + \iota_i\mu_i) = 2\,\mathrm{id}_{\Lambda E}\,.$$

The remaining basis pairs satisfy the Clifford relation by the displayed anticommutation rules and $\theta^2 = 1$. Restricting γ to W gives the D_n model on the same space.

For $U = V, W$ or $u^\perp$, define maps $\varphi : \Lambda^2 U \to \mathfrak{so}(U)$ and $\rho_S : \mathfrak{so}(U) \to \mathfrak{gl}(\Lambda E)$ by

$$\varphi(x\wedge y)z = q(y,z)x - q(x,z)y,$$
$$\rho_S\big(\varphi(x\wedge y)\big) = \frac{1}{4}\big(\gamma(x)\gamma(y) - \gamma(y)\gamma(x)\big).$$

By Lecture 24, φ is an isomorphism and ρ_S is a representation with $[\rho_S(X), \gamma(z)] = \gamma(Xz)$. As in Lecture 24, the subscript names the action on S, because the bare letter ρ is reserved for the Weyl vector.

The conversion table at the end of Lecture 24 compares the normalisations of Lectures 23 to 26. The row of this lecture has the pairing $q(e_j, f_k) = \delta_{jk}$, the isotropic pair $e_j, f_j = \frac{1}{\sqrt{2}}(x_j \pm i x_{n+j})$, the skew map $\varphi(x \wedge y) = A_{xy}$, which is $\frac{1}{2}M_{x,y}$ in the notation of Lecture 24, and the Clifford map with $\gamma(e_j) = \sqrt{2}\,\mu_j$ and $\gamma(f_j) = \sqrt{2}\,\iota_j$. The factor $\sqrt{2}$ is needed because this pairing is twice that of Lecture 24. The orthonormal vectors x_j serve only for this comparison, and this lecture does not use them. Here E is the space called W in Lecture 24, while W is the span of the e_i and f_i.

The spin module of $\mathfrak{so}(V)$ is $S = \Lambda E$. The half-spin modules of $\mathfrak{so}(W)$ are $\Lambda^{\mathrm{ev}}E$ and $\Lambda^{\mathrm{odd}}E$. For $x, y \in W$, the operator $\varphi_V(x \wedge y)$ kills e_0 and restricts to $\varphi_W(x \wedge y)$. The two elements act on ΛE by the same Clifford element. The same holds for $u^{\perp} \subset W$.

Weights. On basis vectors,

$$\varphi(e_i \wedge f_i)e_i = e_i, \qquad \varphi(e_i \wedge f_i)f_i = -f_i,$$

and $\varphi(e_i \wedge f_i)$ kills the other basis vectors. So $\varphi(e_i \wedge f_i) = H_t$ with t the ith unit vector. Its spin action follows from the definition of ρ_S and from $\iota_i\mu_i = 1 - \mu_i\iota_i$:

$$\rho_S\big(\varphi(e_i \wedge f_i)\big) = \frac{1}{4} \cdot 2\,(\mu_i\iota_i - \iota_i\mu_i) = \frac{1}{2}\big(\mu_i\iota_i - (1 - \mu_i\iota_i)\big) = \mu_i\iota_i - \frac{1}{2}.$$

For $I \subseteq \{1, \dots, n\}$ let e_I be the wedge product of the e_i with $i \in I$, in increasing order. Put $s_i(I) = 1$ if $i \in I$, and $s_i(I) = -1$ otherwise. Since $H_t = \sum_i t_i \varphi(e_i \wedge f_i)$, and $\mu_i\iota_i e_I$ is e_I if $i \in I$ and 0 otherwise,

$$\rho_S(H_t)e_I = \frac{1}{2}\sum_{i=1}^{n} s_i(I)\, t_i\, e_I, \qquad \mathrm{wt}(e_I) = \frac{1}{2}\big(s_1(I), \dots, s_n(I)\big).$$

Lemma 26.5 (One dominant weight)**.** *Let $\mathfrak{k}$ be semisimple and M a finite-dimensional $\mathfrak{k}$-module. If M has exactly one dominant weight λ, and $\dim M_\lambda = 1$, then $M \cong L(\lambda)$.*

Proof. By complete reducibility (Lecture 5), M is a direct sum of irreducible submodules. By Lecture 12, each summand has a dominant highest weight, which is a weight of M. So every summand has highest weight λ and contributes to M_λ. Since $\dim M_\lambda = 1$, there is one summand. ∎

Proposition 26.6 (Spinor from B_n to D_n · *[P]*)**.** *Let $n \geq 2$ and use Embedding I. Then*

$$S|_{\mathfrak{so}_{2n}} = \Lambda^{\mathrm{ev}}E \oplus \Lambda^{\mathrm{odd}}E.$$

The summand containing $e_1 \wedge \cdots \wedge e_n$ is $L(\omega_n)$. The other summand is $L(\omega_{n-1})$.

Proof. Submodules. The algebra $\mathfrak{so}(W)$ is spanned by $\varphi(x \wedge y)$ with $x, y \in W$. Each of $\gamma(x)$ and $\gamma(y)$ is a linear combination of the operators ι_i and μ_i, so

$$\gamma(x)\Lambda^k E \subseteq \Lambda^{k-1}E \oplus \Lambda^{k+1}E.$$

Hence $\gamma(x)\gamma(y)$ and $\rho_S(\varphi(x \wedge y))$ preserve the parity of the degree.

Highest weights. The number of entries $-\frac{1}{2}$ in $\mathrm{wt}(e_I)$ is $n - |I|$, so within each half the number of minus signs has a fixed parity. A D_n-dominant weight with all $|\lambda_i| = \frac{1}{2}$ has

$\lambda_1 = \cdots = \lambda_{n-1} = \frac{1}{2}$ and $\lambda_n = \pm\frac{1}{2}$. Thus the half containing $e_{\{1,\dots,n\}}$ has the single dominant weight ω_n. The other half contains $e_{\{1,\dots,n-1\}}$, of opposite parity, and has the single dominant weight ω_{n-1}. All weight spaces are one-dimensional, since different subsets I give different weights. Lemma 26.5 finishes the proof. ∎

Proposition 26.7 (Half-spinors from D_n to B_{n-1} · *[P]*)**.** *Let $n \geq 2$, use Embedding II, and let S' be the spin module of $\mathfrak{so}(u^\perp) \cong \mathfrak{so}_{2n-1}$.*

(a) The operator $\gamma(u)$ restricts to an isomorphism of $\mathfrak{so}(u^\perp)$-modules $\Lambda^{\mathrm{ev}}E \to \Lambda^{\mathrm{odd}}E$.

(b) Each half-spin module restricts to $S' \cong L(\omega_{n-1})$.

Proof. (a) For $x \in u^\perp$ we have $q(u,x) = 0$, so $\gamma(u)\gamma(x) = -\gamma(x)\gamma(u)$. For $x, y \in u^\perp$,

$$\gamma(u)\gamma(x)\gamma(y) = -\gamma(x)\gamma(u)\gamma(y) = \gamma(x)\gamma(y)\gamma(u).$$

Thus $\gamma(u)$ commutes with every $\rho_S(\varphi(x \wedge y))$. By Lemma 26.1 and Lecture 24, the elements $\varphi(x \wedge y)$ span $\mathfrak{so}(u^\perp)$. Next, by the definition of γ and the Clifford relation,

$$\gamma(u) = \sqrt{2}\,(\mu_n + \iota_n), \qquad \gamma(u)^2 = q(u,u) = 2.$$

The first formula shows that $\gamma(u)$ changes parity. The second shows that it is invertible. So $\gamma(u)$ maps each half isomorphically onto the other.

(b) By § 26.1, the restricted weight of e_I is $\frac{1}{2}(s_1(I), \dots, s_{n-1}(I))$. A sign vector in $\{\pm 1\}^{n-1}$ comes from exactly two subsets,

$$I_0 \subseteq \{1, \dots, n-1\} \quad \text{and} \quad I_0 \cup \{n\},$$

of opposite parity. So each half has the weights $\frac{1}{2}(\pm 1, \dots, \pm 1)$, each once. The only B_{n-1}-dominant one is $\omega_{n-1} = \frac{1}{2}(1, \dots, 1)$. Lemma 26.5 gives $L(\omega_{n-1})$. The weight computation above, with $n-1$ in place of n, shows that S' has the same description. ∎

26.4 Worked branching from B_3 to B_2

Take $n = 3$. Use the eight wedge vectors and their weights from the $\mathfrak{so}_6$ example of Lecture 24. For B_3, the only dominant weight is $\frac{1}{2}(1,1,1)$, so $S \cong L(\omega_3)$. For D_3, the odd half has dominant weight $\omega_3 = \frac{1}{2}(1,1,1)$ and the even half has $\omega_2 = \frac{1}{2}(1,1,-1)$. Theorem 26.2 agrees:

$$\frac{1}{2} \geq \mu_1 \geq \frac{1}{2} \geq \mu_2 \geq \frac{1}{2} \geq |\mu_3| \quad \Longrightarrow \quad \mu = \frac{1}{2}(1,1,\pm 1).$$

The dimension count is $8 = 4 + 4$.

Now restrict to B_2 with $u = e_3 + f_3$. Deleting the third coordinate gives the four weights $\frac{1}{2}(\pm 1, \pm 1)$ in each half. Theorem 26.3 agrees. For both halves,

$$\frac{1}{2} \geq \mu_1 \geq \frac{1}{2} \geq \mu_2 \geq \left|\pm\frac{1}{2}\right| \quad \Longrightarrow \quad \mu = (\frac{1}{2}, \frac{1}{2}).$$

Figure 26.1 shows the whole chain. Each single step is multiplicity-free. The two paths end at the same module, so the two-step restriction contains it twice.

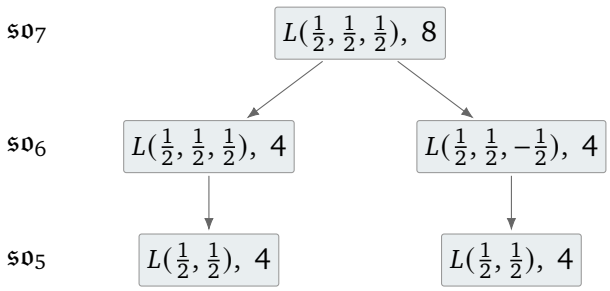


Figure 26.1: The restriction of the B_3 spin module along $\mathfrak{so}_7 \supset \mathfrak{so}_6 \supset \mathfrak{so}_5$. Each box gives a highest weight in ε-coordinates and a dimension, and each row sums to 8.

The intertwiner $\gamma(u) = \sqrt{2}(\mu_3 + \iota_3)$ acts on the even half by

$$\begin{aligned}
\gamma(u)\,1 &= \sqrt{2}\,e_3,\\
\gamma(u)(e_1 \wedge e_2) &= \sqrt{2}\,e_3 \wedge e_1 \wedge e_2 = \sqrt{2}\,e_1 \wedge e_2 \wedge e_3,\\
\gamma(u)(e_1 \wedge e_3) &= \sqrt{2}\,\iota_3(e_1 \wedge e_3) = -\sqrt{2}\,e_1,\\
\gamma(u)(e_2 \wedge e_3) &= -\sqrt{2}\,e_2.
\end{aligned}$$

Each line joins two vectors with the same B_2 weight.

26.5 Symplectic comparison

Let $\mathbb{C}^4$ have ordered symplectic basis (e_1, e_2, f_1, f_2). The symplectic form ω satisfies

$$\omega(e_i, f_j) = \delta_{ij}, \qquad \omega(e_i, e_j) = \omega(f_i, f_j) = 0.$$

The diagonal Cartan subalgebra consists of operators with matrices

$$\operatorname{diag}(t_1, t_2, -t_1, -t_2)$$

in this basis. The roots are $\pm\varepsilon_1 \pm \varepsilon_2$ (short) and $\pm 2\varepsilon_i$ (long). Under the isomorphism $B_2 \cong C_2$, long roots correspond to long roots. In the Bourbaki numbering of the nodes, the two Dynkin labels are exchanged.

Short-root embedding. For $\alpha = \varepsilon_1 - \varepsilon_2$, the root vector E_α acts by

$$E_\alpha e_2 = e_1, \qquad E_\alpha f_1 = -f_2,$$

and kills e_1 and f_2. The coroot h_α has the matrix

$$D(h_\alpha) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Hence $\mathbb{C}^4 \cong V_1 \oplus V_1$, with summands $\operatorname{span}(e_1, e_2)$ and $\operatorname{span}(f_2, f_1)$.

Long-root embedding. For $\beta = 2\varepsilon_1$, the block $\mathfrak{sp}_2$ acts on $\operatorname{span}(e_1, f_1)$ and kills e_2 and f_2. The coroot h_β has the matrix

$$D(h_\beta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Hence $\mathbb{C}^4 \cong V_1 \oplus V_0 \oplus V_0$.

The same two answers appear in the orthogonal model, in other coordinates. We keep the two coordinate systems apart. Above, (a, b) means $a\varepsilon_1 + b\varepsilon_2$ in the symplectic coordinates. Now use the orthogonal model of § 26.1 with $n = 2$, so that e_1, f_1, e_0 are vectors of $\mathbb{C}^5$. Write ε'_1 and ε'_2 for its coordinates, and let (x, y) mean $x\varepsilon'_1 + y\varepsilon'_2$. Lecture 21 matches the two systems by $\varepsilon'_1 = \varepsilon_1 + \varepsilon_2$ and $\varepsilon'_2 = \varepsilon_1 - \varepsilon_2$. Hence

$$(x, y) = \Big(\frac{a+b}{2}, \frac{a-b}{2}\Big).$$

The B_2 spin module has highest weight ω_2 and dimension 4 (Exercise 24.2). The isomorphism $\mathfrak{sp}_4 \cong \mathfrak{so}_5$ of Lecture 21 identifies it with the defining module $\mathbb{C}^4$ of $\mathfrak{sp}_4$. The highest weight $(a, b) = (1, 0)$ of $\mathbb{C}^4$ becomes $(x, y) = (\frac{1}{2}, \frac{1}{2}) = \omega_2$, and all four weights of $\mathbb{C}^4$ become the weights $\frac{1}{2}(\pm 1, \pm 1)$.

Short root of B_2. The short root ε'_1 gives the standard $\mathfrak{so}_3$ on $\operatorname{span}(e_1, f_1, e_0)$. Its coroot $2H_{(1,0)}$ takes the value $2x$ on the weight (x, y). On the four spin weights these values are

$$1,\ 1,\ -1,\ -1, \qquad \text{which gives } V_1 \oplus V_1.$$

Long root of B_2. A long root $\varepsilon'_1 \pm \varepsilon'_2$ gives one factor of $\mathfrak{so}_4$. Its coroot $H_{(1,\pm 1)}$ takes the value $x \pm y$ on the weight (x, y). On the four spin weights these values are

$$1,\ 0,\ 0,\ -1, \qquad \text{which gives } V_1 \oplus 2V_0.$$

This agrees with the half-spinors $(V_1 \boxtimes V_0) \oplus (V_0 \boxtimes V_1)$, restricted to one factor. So "a root $\mathfrak{sl}_2$" does not determine the answer.

Theorem 26.8 (Symplectic branching · *[C]*)**.** *Use Lecture 21's symplectic model for $n \geq 2$. Let $\mathfrak{sp}_{2n-2} \subset \mathfrak{sp}_{2n}$ consist of the elements killing e_n and f_n, acting on the remaining $n-1$ pairs. Its diagonal Cartan subalgebra has $t_n = 0$, so weights restrict by deleting the last coordinate. Let $\lambda_1 \geq \cdots \geq \lambda_n \geq 0$ and $\mu_1 \geq \cdots \geq \mu_{n-1} \geq 0$ be integers. The multiplicity $m(\lambda, \mu)$ of $L(\mu)$ in $L(\lambda)|_{\mathfrak{sp}_{2n-2}}$ is nonzero if and only if*

$$\lambda_j \geq \mu_j \geq \lambda_{j+2} \quad \text{for } 1 \leq j \leq n-1, \quad \text{where} \quad \lambda_{n+1} = 0.$$

In that case, let $x_1 \geq y_1 \geq \cdots \geq x_n \geq y_n$ be the nonincreasing rearrangement of $\lambda_1, \ldots, \lambda_n, \mu_1, \ldots, \mu_{n-1}, 0$. We have

$$m(\lambda, \mu) = \prod_{j=1}^{n} (x_j - y_j + 1).$$

We import this theorem as the Lie algebra form of Goodman–Wallach, Theorem 8.1.5 (Section 8.1.1, p. 365), for the specified block embedding. The rule is *not* multiplicity-free.

For $n = 2$ the subalgebra is the long-root block, and $L(\mu_1) = V_{\mu_1}$. For the adjoint module $\lambda = (2, 0)$, the condition is $2 \geq \mu_1 \geq 0$. The rearrangements of $\{2, 0, \mu_1, 0\}$ give

$$\begin{aligned} \mu_1 = 2 &: \ (2, 2, 0, 0), \quad m = (2-2+1)(0-0+1) = 1, \\ \mu_1 = 1 &: \ (2, 1, 0, 0), \quad m = (2-1+1)(0-0+1) = 2, \\ \mu_1 = 0 &: \ (2, 0, 0, 0), \quad m = (2-0+1)(0-0+1) = 3. \end{aligned}$$

The table collects this and the defining module:

λ	dim	μ_1 and $m(\lambda, \mu)$	restriction	check
$(1, 0)$	4	1: 1, 0: 2	$V_1 \oplus 2V_0$	$2 + 2$
$(2, 0)$	10	2: 1, 1: 2, 0: 3	$V_2 \oplus 2V_1 \oplus 3V_0$	$3 + 4 + 3$

26.5.1 Minuscule shortcuts

This subsection is a sketch [S]. Call $L(\lambda)$ *minuscule* if its weights form one Weyl orbit. Multiplicities are Weyl-invariant and $\dim L(\lambda)_\lambda = 1$, so every multiplicity is 1 and the character is the orbit sum. The spin and half-spin modules are minuscule by their weight lists in Lecture 24 and the Weyl actions of Lecture 9. The defining $\mathfrak{sp}_{2n}$-module is minuscule by its weight list in Lecture 21. The vector module of B_n and $L(\omega_2)$ of $\mathfrak{sp}_4$ are not, since their zero weight is a separate orbit.

Utility and limits. Restrict the orbit sum and subtract irreducible characters in highest-weight order, as in Lecture 17. The embedding determines this restriction. Multiplicities can appear, as for $\mathbb{C}^4 \to V_1 \oplus V_1$. The method gives characters; the operator $\gamma(u)$ supplies an intertwiner. The general classification of minuscule weights is omitted.

Self-checks

(1) Restrict the vector module of $\mathfrak{so}_7$ to $\mathfrak{so}_6$ and then to $\mathfrak{so}_5$.

(2) Which $\mathfrak{so}_6$ highest weights occur in the adjoint module $L(1,1,0)$ of $\mathfrak{so}_7$?

(3) What is the restriction of the $\mathfrak{sp}_4$-module $\mathbb{C}^4$ to a root $\mathfrak{sl}_2$?

Answers. (1) The two steps are

$$B_3 \to D_3 : \ (1,0,0) \mapsto (1,0,0) + (0,0,0), \quad 7 = 6 + 1,$$
$$D_3 \to B_2 : \ (1,0,0) \mapsto (1,0) + (0,0), \quad 6 = 5 + 1,$$

and the trivial summand restricts to the trivial module $(0,0)$, so $7 = 5 + 1 + 1$. (2) The rule $1 \geq \mu_1 \geq 1 \geq \mu_2 \geq 0 \geq |\mu_3|$ gives $(1,1,0)$ and $(1,0,0)$, and $21 = 15 + 6$. (3) The question is incomplete. A short root gives $V_1 \oplus V_1$, with coroot eigenvalues $1, -1, -1, 1$. A long root gives $V_1 \oplus 2V_0$, with eigenvalues $1, 0, -1, 0$.

Summary

Orthogonal restriction along Embeddings I and II, the stabilisers of nonisotropic vectors, is multiplicity-free. The highest weights of the summands interlace with the given one and have its type.

The spin cases follow from the exterior model. Parity splits the B_n spinor, and $\gamma(u)$ identifies the two D_n half-spinors over B_{n-1}.

Symplectic restriction to the block $\mathfrak{sp}_{2n-2}$ has a product formula with multiplicities above one. Every comparison begins by naming the embedding.

Reading

Goodman and Wallach, *Symmetry, Representations, and Invariants* (2009), Section 8.1.1, Theorems 8.1.3–8.1.5, pp. 364–365. Fulton and Harris, *Representation Theory: A First Course*, Lectures 18–20 on orthogonal Lie algebras and spin representations, and §25.3 on branching rules. Exercise Sheet 26 gives further restrictions and proofs.